

Practical 9:

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#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <stdlib.h>

#define BUFFER_SIZE 4096

int main(int argc, char *argv[])
{
    int src_fd, dest_fd;
    char buffer[BUFFER_SIZE];
    ssize_t bytes_read, bytes_written;

    if (argc != 3)
    {
        printf("Usage: %s <source> <destination>\n", argv[0]);
        return 1;
    }

    // Open source file for reading
    src_fd = open(argv[1], O_RDONLY);

    if (src_fd == -1)
    {
        perror("open source");
        return 1;
    }

    // Create/open destination file for writing
    dest_fd = open(argv[2], O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (dest_fd == -1)
    {
        perror("open destination");
        close(src_fd);
        return 1;
    }
}
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        return 1;
    }

    // Copy data
    while ((bytes_read = read(src_fd, buffer, BUFFER_SIZE)) > 0)
    {
        bytes_written = write(dest_fd, buffer, bytes_read);

        if (bytes_written != bytes_read)
        {
            perror("write");
            close(src_fd);
            close(dest_fd);
            return 1;
        }
    }

    if (bytes_read == -1)
    {
        perror("read");
    }

    // Demonstrate lseek()
    if (lseek(src_fd, 0, SEEK_SET) == -1)
    {
        perror("lseek");
    }

    close(src_fd);
    close(dest_fd);

    printf("File copied successfully using low-level system calls.\n");

    return 0;
}

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```
#include <stdio.h>
#include <stdlib.h>

#define BUFFER_SIZE 4096

int main(int argc, char *argv[])
{
    FILE *src;
    FILE *dest;
    char buffer[BUFFER_SIZE];
    size_t bytes_read;

    if (argc != 3)
    {
        printf("Usage: %s <source> <destination>\n", argv[0]);
        return 1;
    }

    src = fopen(argv[1], "rb");

    if (src == NULL)
    {
        perror("fopen source");
        return 1;
    }

    dest = fopen(argv[2], "wb");

    if (dest == NULL)
    {
        perror("fopen destination");
        fclose(src);
        return 1;
    }

    while ((bytes_read = fread(buffer, 1, BUFFER_SIZE, src)) > 0)
    {

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    }

    dest = fopen(argv[2], "wb");

    if (dest == NULL)
    {
        perror("fopen destination");
        fclose(src);
        return 1;
    }

    while ((bytes_read = fread(buffer, 1, BUFFER_SIZE, src)) > 0)
    {
        if (fwrite(buffer, 1, bytes_read, dest) != bytes_read)
        {
            perror("fwrite");
            fclose(src);
            fclose(dest);
            return 1;
        }
    }

    fclose(src);
    fclose(dest);

    printf("File copied successfully using standard library functions.\n");

    return 0;
}
|

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bhanu@DESKTOP-DE32HQ8:~$ cd ~
mkdir -p file_io_lab
cd file_io_lab
bhanu@DESKTOP-DE32HQ8:~/file_io_lab$ pwd
/home/bhanu/file_io_lab
bhanu@DESKTOP-DE32HQ8:~/file_io_lab$ nano source.txt
bhanu@DESKTOP-DE32HQ8:~/file_io_lab$ nano lowlevel_copy.c
bhanu@DESKTOP-DE32HQ8:~/file_io_lab$ gcc -Wall -Wextra -o lowlevel_copy lowlevel_copy.c
bhanu@DESKTOP-DE32HQ8:~/file_io_lab$ ./lowlevel_copy source.txt copied_lowlevel.txt
File copied successfully using low-level system calls.
bhanu@DESKTOP-DE32HQ8:~/file_io_lab$ cat copied_lowlevel.txt
This is a test file for Linux file I/O.
We are using open, read, write, lseek and close system calls.
This program copies data from one file to another.
bhanu@DESKTOP-DE32HQ8:~/file_io_lab$ cat copied_lowlevel.txt
This is a test file for Linux file I/O.
We are using open, read, write, lseek and close system calls.
This program copies data from one file to another.
bhanu@DESKTOP-DE32HQ8:~/file_io_lab$ ls -l source.txt copied_lowlevel.txt
-rw-r--r-- 1 bhanu bhanu 153 Oct  2 16:39 copied_lowlevel.txt
-rw-r--r-- 1 bhanu bhanu 153 Oct  2 16:39 source.txt
bhanu@DESKTOP-DE32HQ8:~/file_io_lab$ nano library_copy.c
bhanu@DESKTOP-DE32HQ8:~/file_io_lab$ gcc -Wall -Wextra -o library_copy library_copy.c
bhanu@DESKTOP-DE32HQ8:~/file_io_lab$ ./library_copy source.txt copied_library.txt
File copied successfully using standard library functions.
bhanu@DESKTOP-DE32HQ8:~/file_io_lab$ diff source.txt copied_library.txt
bhanu@DESKTOP-DE32HQ8:~/file_io_lab$ diff source.txt copied_library.txt
bhanu@DESKTOP-DE32HQ8:~/file_io_lab$ time ./lowlevel_copy largefile.bin lowlevel.bin
open source: No such file or directory

real    0m0.001s
user    0m0.000s
sys     0m0.000s
bhanu@DESKTOP-DE32HQ8:~/file_io_lab$ time ./library_copy largefile.bin library.bin
fopen source: No such file or directory

real    0m0.001s
user    0m0.000s
sys     0m0.000s
bhanu@DESKTOP-DE32HQ8:~/file_io_lab$ time ./lowlevel_copy largefile.bin lowlevel.bin
open source: No such file or directory

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bhanu@DESKTOP-DE32HQ8:~/file_io_lab$ time ./lowlevel_copy largefile.bin lowlevel.bin
open source: No such file or directory

real    0m0.001s
user    0m0.001s
sys     0m0.000s
bhanu@DESKTOP-DE32HQ8:~/file_io_lab$ time ./library_copy largefile.bin library.bin
fopen source: No such file or directory

real    0m0.001s
user    0m0.001s
sys     0m0.000s
bhanu@DESKTOP-DE32HQ8:~/file_io_lab$ |

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